Asymmetric Information

18

The buyer needs a hundred eyes, the seller not one. —George Herbert (1651)

CHALLENGE

Dying to Work

In part because of the differing amounts that firms invest in safety, jobs in some firms are more dangerous than in others. In 2010, the United States had a particularly bad year for major disasters: A blast at a Washington state refinery killed 7 workers, 11 workers died when BP's Deepwater Horizon oil rig exploded in the Gulf of Mexico, and 29 coal miners died in a disaster at Massey Energy's West Virginia mine explosion. These disasters are only the tip of the iceberg as thousands of workers are killed on the job every year—4,585 in 2013. Major disasters have occurred in other countries. An apparel factory collapse in Bangladesh killed 1,129 workers in 2013. A warehouse explosion in the port of Tianjin, China in 2015 killed at least 39 workers.

Rational people who fear danger agree to work in dangerous jobs only if those jobs pay a sufficiently higher wage than less-risky alternative jobs.[1] Workers receive such *compensating wage differentials* in industries and occupations that government statistics show are relatively risky.

Injury rates vary dramatically by industry, ranging from only 0.9 fatal injuries per 100,000 workers in the financial services industry to 9.7 in construction and 12.4 in mining. Some occupations are particularly dangerous. Logging had a fatal injury rate of 91.3 per 100,000 workers; fishing 75.0; agriculture, 22.9; and driving a truck, 23.6. On the other hand, safe occupations include sales, 1.6 and educational services, 0.7 (although students sometimes risk dying of boredom).[2]

If people are rational and fear danger, they agree to work in a dangerous job only if that job pays a sufficiently higher wage than less-risky alternative jobs. Economists find that workers receive *compensating wage differentials* in industries and occupations that government statistics show are relatively risky.

However, if workers are unaware of the greater risks at certain firms within an industry, they may not receive compensating wage differentials from more dangerous employers within that industry. Workers are likely to have a sense of the risks associated with an industry:

[1]In 2012, Redhook Brewery started selling a special beer in memory of an employee who was killed when a keg exploded, with the proceeds going to his family (**http://boston.cbslocal.com**, July 20, 2012).

[2]Government statistics also tell us that males have an accident rate, 6.0, that is an order of magnitude greater than females, 0.7. Some of this difference is due to different occupations and some to different attitudes toward risk. How many women are injured after saying, "Hey! Watch this!"?

Everyone knows that mining is relatively risky—but they do not know which mining companies are particularly risky until a major accident occurs. For example, in the past decade, 54 coal miners have been killed in Massey mines, a much higher rate than at other firms' mines, yet these workers apparently did not receive higher pay than workers at other mining firms.[3]

One justification often given for government intervention is that firms have more information than workers do about job safety at their plants. Prospective employees often do not know the injury rates at individual firms but may know the *average* injury rate over an entire industry, in part because governments report such data. Does such a situation cause firms to underinvest in safety? Can government intervention overcome such safety problems?

So far, we've examined models in which everyone is equally knowledgeable or equally ignorant: They have *symmetric information*. In the competitive model, everyone knows the relevant facts. In uncertainty models in Chapter 16, the companies that sell insurance and the people who buy it may be equally uncertain about future events. In contrast, in this chapter's models, people have **asymmetric information**: One party to a transaction has relevant information that another party lacks. For example, the seller knows the quality of a product and the buyer does not.

Two important types of asymmetric information are *hidden characteristics* and *hidden actions*. A **hidden characteristic** is a fact about a person or thing that one party to a transaction knows but that is unknown to other parties. For example, the owner of a property may possess extensive information about the mineral composition of the land that is unknown to a mining company that is considering buying the land.

A **hidden action** is an act by one party to a transaction that is unknown by the other parties. An example is a firm's manager using a company jet for personal use without the owners' knowledge.

When both parties to a transaction have equal information or equally limited information, neither has an advantage over the other. However, asymmetric information leads to **opportunistic behavior**, where one party takes economic advantage of another when circumstances permit. Such *opportunistic behavior* due to asymmetric information leads to market failures, and destroys many desirable properties of competitive markets.

Two problems of opportunistic behavior arise from asymmetric information. One—*adverse selection*—is due to hidden characteristics, while the other—*moral hazard*—is associated with hidden actions. The problem of **adverse selection** arises when one party to a transaction possesses information about a hidden characteristic that is unknown to other parties and takes economic advantage of this information. For example, if a roadside vendor sells a box of oranges to a passing motorist and only the vendor knows that the oranges are of low quality, the vendor may allege that the oranges are of high quality and charge a premium price for them. That is, the seller seeks to benefit from an informational asymmetry due to a hidden characteristic, the quality of the oranges. If potential buyers worry about such opportunistic behavior, they may be willing to pay only low prices or may forgo purchasing the oranges entirely.

[3]The U.S. Mine Safety and Health Administration issued Massey 124 safety-related citations in 2010 prior to the April 2010 accident at Massey's Upper Big Branch mine in West Virginia that killed 29 workers. Massey had 515 violations in 2009. Mine Safety and Health Administration safety officials concluded in 2011 that Massey could have prevented the 2010 explosion that took 29 lives. Justice Department officials brought a criminal indictment against the former head of safety at the mine.

The primary problem arising from hidden action is **moral hazard**, which occurs when an informed party takes an action that the other party cannot observe and that harms the less-informed party. If you pay a mechanic by the hour to fix your car, and you do not actually watch the repairs, then the time spent by the mechanic on your car is a hidden action. Moral hazard occurs if the mechanic bills you for excessive hours.

This chapter focuses on adverse selection and unobserved characteristics. Adverse selection often leads to markets in which some desirable transactions do not take place or even the market as a whole cannot exist. We also discuss methods that sometimes solve adverse selection problems. Chapter 19 concentrates on moral hazard problems due to unobserved actions and on the use of contracts to deal with them.

In this chapter, we examine four main topics

1. **Adverse Selection.** Adverse selection may prevent desirable transactions from occurring, possibly eliminating a market.
2. **Reducing Adverse Selection.** Government regulations, contracts between involved parties, and means of equalizing information may reduce or eliminate the harms from adverse selection.
3. **Market Power from Price Ignorance.** Firms gain market power from consumers' ignorance about the price that each firm charges.
4. **Problems Arising from Ignorance When Hiring.** Attempts to eliminate information asymmetries in hiring may raise or lower social welfare.

18.1 Adverse Selection

One of the most important problems associated with adverse selection is that consumers may not make purchases to avoid being exploited by better-informed sellers. As a result, not all desirable transactions occur, and potential consumer and producer surplus is lost. Indeed, in the extreme case, adverse selection may prevent a market from operating at all. We illustrate this idea using two important examples of adverse selection problems: insurance and products of unknown quality.

Insurance Markets

Hidden characteristics and adverse selection are very important in the insurance industry. Were a health insurance company to provide fair insurance by charging everyone a rate for insurance equal to the average cost of health care for the entire population, the company would lose money due to adverse selection. Many unhealthy people—people who expect to incur health care costs that are higher than average—would view this insurance as a good deal and would buy it. In contrast, unless they were very risk averse, healthy people would not buy it because the premiums exceed their expected health care costs. Given that a disproportionately large share of unhealthy people would buy the insurance, the market for health insurance would exhibit adverse selection, and the insurance company's average cost of medical care for covered people would exceed the population's average.

Adverse selection results in an inefficient market outcome because the sum of producer and consumer surplus is not maximized. The loss of potential surplus occurs because some potentially beneficial sales of insurance to relatively healthy individuals do not occur. These consumers are willing to buy insurance at a lower rate that is closer to the fair rate for them given their superior health. The insurance company is willing to offer such low rates only if it is sure that these individuals are healthy.

Products of Unknown Quality

Anagram for General Motors: or great lemons.

Adverse selection often arises because sellers of a product have better information about the product's quality—a hidden characteristic—than do buyers. Used cars that appear to be identical on the outside often differ substantially in the number of repairs they will need. Some cars—*lemons*—are cursed. They have a variety of insidious problems that become apparent to the owner only after driving the car for a while. Thus, the owner knows from experience if a used car is a lemon, but a potential buyer does not.

If buyers have the same information as sellers, no adverse selection problem arises. However, when sellers have more information than buyers, adverse selection may drive high-quality products out of the market (Akerlof, 1970). Why? Used car buyers worry that a used car might be a lemon. As a result, they will not pay as high a price as they would if they knew the car was of good quality. They will only buy if the price is low enough to reflect the possibility of getting a lemon. Given that sellers of excellent used cars do not want to sell their cars for that low a price, they do not enter the market. Adverse selection has driven the high-quality cars out of the market, leaving only the lemons.

In the following example, we assume that sellers cannot alter the quality of their used cars and that the number of potential used car buyers is large. All potential buyers are willing to pay \$4,000 for a lemon and \$8,000 for a good used car: The demand curve for lemons, D^L, is horizontal at \$4,000 in panel a of Figure 18.1, and the demand curve for good cars, D^G, is horizontal at \$8,000 in panel b.

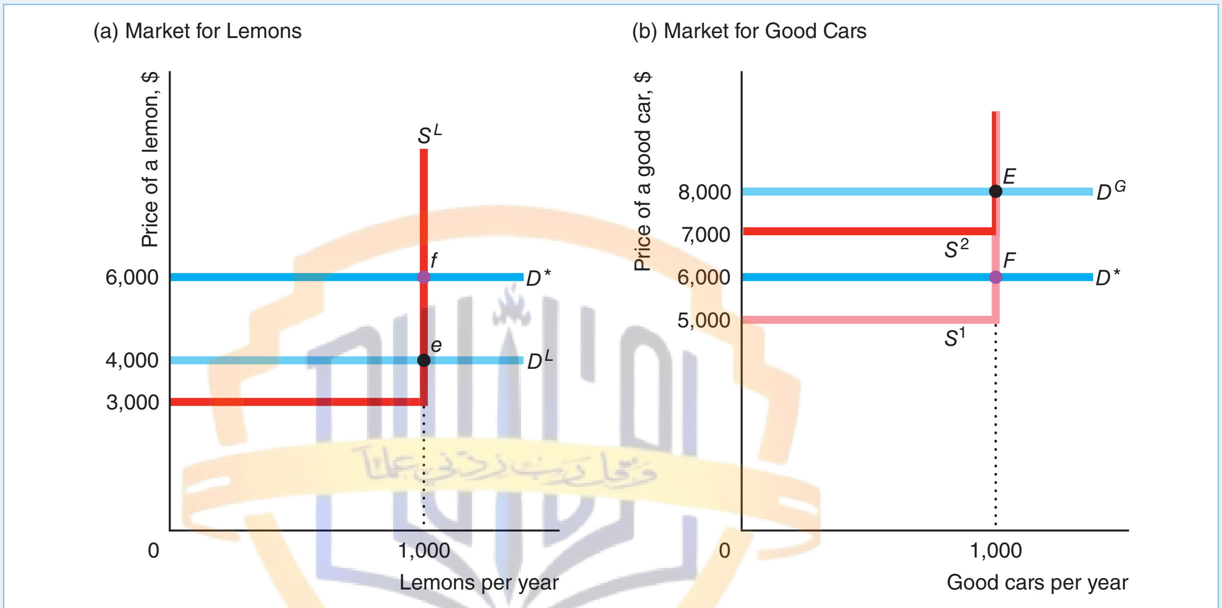
Although the number of potential buyers is virtually unlimited, only 1,000 owners of lemons and 1,000 owners of good cars are willing to sell. The *reservation price* of lemon owners—the lowest price at which they will sell their cars—is \$3,000. Consequently, the supply curve for lemons, S^L in panel a, is horizontal at \$3,000 up to 1,000 cars, where it becomes vertical (no more cars are for sale at any price). The reservation price of owners of high-quality used cars is v, which is less than \$8,000. Panel b shows two possible values of v. If $v = \$5,000$, the supply curve for good cars, S^1, is horizontal at \$5,000 up to 1,000 cars and then becomes vertical. If $v = \$7,000$, the supply curve is S^2.

Market Equilibrium with Symmetric Information. If sellers and buyers know the quality of all the used cars before any sales take place (they have full, symmetric information), all 2,000 cars are sold, and the good cars sell for more than the lemons. In panel a of Figure 18.1, the intersection of the lemons demand curve D^L and the lemons supply curve S^L determines the equilibrium at e in the lemons market, where 1,000 lemons sell for \$4,000 each. Regardless of whether the supply curve for good cars is S^1 or S^2 in panel b, the equilibrium in the good-car market is E, where 1,000 good cars sell for \$8,000 each.

Figure 18.1 Markets for Lemons and Good Cars

MyEconLab Video

If everyone has full information, the equilibrium in the lemons market is *e* (1,000 cars sold for \$4,000 each), and the equilibrium in the good-car market is *E* (1,000 cars sold for \$8,000 each). If buyers can't tell quality before buying but assume that equal numbers of the two types of cars are for sale, their demand in both markets is D^*, which is horizontal at \$6,000. If the good car owners' reservation price is \$5,000, the supply curve for good cars is S^1, and 1,000 good cars (point *F*) and 1,000 lemons (point *f*) sell for \$6,000 each. If their reservation price is \$7,000, the supply curve is S^2. No good cars are sold; 1,000 lemons sell for \$4,000 each (point *e*).



This market is efficient because the goods go to the people who value them the most. All current owners, who value the cars less than the potential buyers, sell their cars.

More generally, all buyers and sellers may have symmetric information by being equally informed or equally uninformed. *All the cars are sold if everyone has the same information.* It does not matter whether they all have full information or all lack information—it's the equality of information that matters. However, *the amount of information they have affects the price at which the cars sell.* With full information, good cars sell for \$8,000 and lemons sell for \$4,000.

If information is symmetric because buyers and sellers are equally ignorant (neither group knows if a car is good or a lemon), all cars sell for the same price. A buyer has an equal chance of buying a lemon or a good car. The expected value of a used car is

$$\left(\tfrac{1}{2} \times \$4{,}000\right) + \left(\tfrac{1}{2} \times \$8{,}000\right) = \$6{,}000.$$

Suppose buyers and sellers are risk neutral (Chapter 16). Buyers are willing to pay \$6,000 for a car of unknown quality. Because sellers cannot distinguish between the cars either, sellers accept this amount and sell all the cars.[4] Thus, this market is efficient because the cars go to people who value them more than their original owners.

[4]Risk-neutral sellers place an expected value of $(\tfrac{1}{2} \times \$3{,}000) + \tfrac{1}{2}v = \$1{,}500 + \tfrac{1}{2}v$ on a car of unknown quality. If $v = 7{,}000$, this expected value is \$1,500 + \$3,500 = \$5,000. If $v = \$5{,}000$ this expected value is only \$4,000. In either case, sellers are happy to sell their cars for \$6,000.

If only lemons were sold, they would sell for \$4,000. The presence of good-quality cars raises the price received by sellers of lemons to \$6,000. Similarly, if only good cars were sold, they would sell for \$8,000. The presence of lemons lowers the price that sellers of good cars receive to \$6,000. Thus, *sellers of good-quality cars are effectively subsidizing sellers of lemons*.

Market Equilibrium with Asymmetric Information. If sellers know the quality but buyers do not, the market may be inefficient: Possibly, owners sell none of the better-quality cars even though buyers value good cars more than the owners do. The equilibrium in this market depends on whether the value that the owners of good cars place on their cars, v, is greater or less than the expected value of buyers, \$6,000. The two possible equilibria are: (1) All cars sell at the average price, or (2) only lemons sell at a price equal to the value that buyers place on lemons.

Initially, let's assume that the sellers of good cars value their cars at $v = \$5{,}000$, which is less than the buyers' expected value of the cars, \$6,000, so transactions can occur. The equilibrium in the good-car market is determined by the intersection of S^1 and D^* at point *F*, where 1,000 good cars sell for \$6,000. Similarly, owners of lemons, who value their cars at only \$3,000, are very happy to sell them for \$6,000. The new equilibrium in the lemons market is *f*.

Thus, all cars sell at the same price. Consequently, here *asymmetric information does not cause an efficiency problem, but it does have equity implications*. Sellers of lemons benefit and sellers of good cars suffer from consumers' inability to distinguish quality. Consumers who buy the good cars get a bargain, and buyers of lemons are left with a sour taste in their mouths.

Now suppose that the sellers of good cars place a value of $v = \$7{,}000$ on their cars, so they are unwilling to sell them for \$6,000. As a result, the *lemons drive good cars out of the market*. Buyers realize that they can buy only lemons at any price less than \$7,000. Consequently, in equilibrium, the 1,000 lemons sell for the expected (and actual) price of \$4,000, and no good cars change hands. This equilibrium is inefficient because high-quality cars remain in the hands of people who value them less than potential buyers do.

In summary, if buyers have less information about product quality than sellers do, the result might be a lemons problem in which high-quality cars do not sell, even though potential buyers value the cars more than their current owners do. If so, the asymmetric information causes a competitive market to lose its desirable efficiency and welfare properties. However, if the information is symmetric, the lemons problem does not occur. That is, if buyers and sellers of used cars know the quality of the cars, each car sells for its true value in a perfectly competitive market. Moreover, if, as with new cars, neither buyers nor sellers can identify lemons, both good cars and lemons sell at a price equal to the expected value rather than at their (unknown) true values.

SOLVED PROBLEM 18.1

MyEconLab Solved Problem

Suppose that everyone in our used-car example is risk neutral; potential car buyers value lemons at \$4,000 and good used cars at \$8,000; the reservation price of lemon owners is \$3,000; and the reservation price of owners of high-quality used cars is \$7,000. The share of current owners who have lemons is θ. (In our previous example, the share was $\theta = \frac{1}{2} = 1{,}000/[1{,}000 + 1{,}000]$). For what values of θ do all the potential sellers sell their used cars? Describe the equilibrium.

Answer

1. *Determine how much buyers are willing to pay if all cars are sold.* Because buyers are risk neutral, if they believe that the probability of getting a lemon is θ, the most they are willing to pay for a car of unknown quality is

$$p = [\$8{,}000 \times (1 - \theta)] + (\$4{,}000 \times \theta) = \$8{,}000 - (\$4{,}000 \times \theta). \quad (18.1)$$

For example, $p = \$6{,}000$ if $\theta = \frac{1}{2}$ and $p = \$7{,}000$ if $\theta = \frac{1}{4}$.

2. *Solve for the values of θ such that all the cars are sold, and describe the equilibrium.* All owners will sell if the market price equals or exceeds their reservation price, \$7,000. Using Equation 18.1, we know that the market (equilibrium) price is \$7,000 or more if a quarter or fewer of the used cars are lemons, $\theta \leq \frac{1}{4}$. Thus, for $\theta \leq \frac{1}{4}$, all the cars are sold at the price given in Equation 18.1.

Lemons Market with Variable Quality

Most firms can adjust their product's quality. If consumers cannot identify high-quality goods before purchase, they pay the same for all goods regardless of quality. Because the price that firms receive for top-quality goods is the same they receive for low-quality items, they do not produce top-quality goods. Such an outcome is inefficient if consumers are willing to pay sufficiently more for top-quality goods.

This unwillingness to produce high-quality products is due to an externality (Chapter 18): *A firm does not completely capture the benefits from raising the quality of its product*. By selling a better product than what other firms offer, a seller raises the average quality in the market, so buyers are willing to pay more for all products. As a result, the high-quality seller shares the benefits from its high-quality product with sellers of low-quality products by raising the average price to all. *The social value of raising the quality*, as reflected by the increased revenues shared by all firms, *is greater than the private value*, which is only the higher revenue received by the firm with the good product.

SOLVED PROBLEM 18.2

MyEconLab Solved Problem

It costs \$10 to produce a low-quality wallet and \$20 to produce a high-quality wallet. Consumers cannot distinguish between the products before purchase, do not make repeat purchases, and value the wallets at the cost of production. The five firms in the market produce 100 wallets each. Each firm produces only high-quality or only low-quality wallets. Consumers pay the expected value of a wallet. Do any of the firms produce high-quality wallets?

Answer

Show that it does not pay for one firm to make high-quality wallets if the other firms make low-quality wallets due to asymmetric information. If all five firms make low-quality wallets, consumers pay \$10 per wallet. If one firm makes high-quality wallets and all the others make low-quality wallets, the expected value per wallet to consumers is

$$\left(\$10 \times \tfrac{4}{5}\right) + \left(\$20 \times \tfrac{1}{5}\right) = \$12.$$

Thus, if one firm raises the quality of its product, all firms benefit because the wallets sell for \$12 instead of \$10. The high-quality firm receives only a fraction of the total benefit from raising quality. It gets $2 extra per high-quality wallet sold, which is less than the extra $10 it costs to make the better wallet. The other firms share the other $8. Because the high-quality firm incurs all the expenses of raising quality, $10 extra per wallet, and reaps only a fraction, $2, of the benefits, it opts not to produce high-quality wallets. Therefore, *due to asymmetric information, the firms do not produce high-quality goods even though consumers are willing to pay for the extra quality.*

18.2 Reducing Adverse Selection

Because adverse selection results from one party exploiting asymmetric information about a hidden characteristic, the two main methods for solving adverse selection problems are to *equalize information* among the parties and to *restrict the ability of the informed party to take advantage of hidden information*. Responses to adverse selection problems increase welfare in some markets, but may do more harm than good in others.

Equalizing Information

Providing information to all parties eliminates adverse selection problems. Either informed or uninformed parties can act to eliminate or reduce informational asymmetries. Three methods for reducing informational asymmetries are:

1. An uninformed party (such as an insurance company) can use **screening** to infer the information possessed by informed parties.
2. An informed party (such as a person seeking to buy health insurance) can use **signaling** to send information to a less-informed party.
3. A third party, such as a firm or a government agency, not directly involved in the transaction may collect information and sell or give it to the uninformed party.

Screening. Uninformed people may eliminate their disadvantage by screening to gather information on the hidden characteristics of informed people. Life insurance companies reduce adverse selection problems by requiring medical exams. Based on this information, a firm may decide not to insure high-risk individuals or to charge them a higher premium as compensation for the extra risk.

Congratulations! You qualify for our auto insurance.

It is costly to collect information on a person's health or dangerous habits such as smoking, drinking, or skydiving. As a result, insurance companies collect information only up to the point at which the marginal benefit from the extra information they gather equals the marginal cost of obtaining it. Over time, insurance companies have increasingly concluded that it pays to collect information about whether individuals exercise, have a family history of dying young, or engage in potentially life-threatening activities.

Consumers can use screening techniques too. For example, a potential customer can screen a used car by test-driving it, by having an objective, trustworthy mechanic examine the car, or by paying a company such as CARFAX to check the history of the repairs on the vehicle. As long as the consumers' costs of securing information are less than the private benefits, they obtain the information, transactions occur, and markets function smoothly.

In some markets, consumers can avoid the adverse selection problem by buying only from a firm that has a *reputation* for providing high-quality goods. For example, consumers know that a used car dealer that expects repeat purchases has a stronger incentive not to sell defective cars than does an individual.

Signaling. An informed party may signal the uninformed party to eliminate adverse selection. However, signals solve the adverse selection problem only when the recipients view them as credible. Smart consumers may place little confidence in a firm's unsubstantiated claims. Do you believe that a used car runs well just because an ad tells you so?

If only high-quality firms find it worthwhile to send a signal, then a signal is credible. Producers of high-quality goods often try to signal to consumers that their products are of better quality than those of their rivals. If consumers believe their signals, these firms can charge higher prices for their goods.

But if the signals are to be effective, they must be credible. For example, a firm may distribute a favorable review of its product by an independent testing agency to try to convince buyers that its product is of high quality. If low-quality firms cannot obtain such a report from a reliable independent testing agency, consumers believe this signal.

A warranty may serve as both a signal and a guarantee. It is less expensive for the manufacturer of a reliable product to offer a warranty than it is for a firm that produces low-quality products. Consequently, if one firm offers a warranty and another does not, then a consumer may infer that the firm with the warranty produces a superior product. Of course, sleazy firms may try to imitate high-quality firms by offering a warranty that they do not intend to honor.

An applicant for life insurance can present an insurance company with a written statement from the doctor as a signal of good health. If only people who believe that they can show that they are better than others want to send a signal, insurance companies may rely upon it. However, an insurance company may not trust such a signal if it is easy for people to find unscrupulous doctors who will report falsely that they are in good health. Here, screening by the insurance company using its own doctors may work better because the information is more credible.

APPLICATION

Discounts for Data

Are you healthy? Your insurance company wants to know. And it wants to keep you healthy. Life insurance companies in Australia, Europe, Singapore, and South Africa use the Internet to allow customers to signal that they're healthy.

When Andrew Thomas swipes his membership card when he arrives at his gym, his South African life insurance company, Vitality, receives instant information. The company checks whether he's still there 30 minutes later by tracking his location using his smartphone. In return for sharing medical and exercise information with his insurance company, Mr. Thomas earns points, which reduce his insurance premium by 9%.

In 2015, John Hancock became the first U.S. life insurance company to introduce a similar program. The company provides customers with Fitbit monitors that automatically upload their activity levels. The most active customers will earn a discount of up to 15% on their life-insurance premium, Amazon gift cards, and half-price stays at Hyatt hotels.

Thus, life insurance companies are helping their customers signal that they're healthy, reducing the adverse selection problem. In addition, the companies are using this information to provide incentives for their customers to lead healthier lives so that their families have to wait longer to collect on their life insurance.

Third-Party Information. In some markets, consumer groups, nonprofit organizations, and government agencies provide buyers with information about the quality of different goods and services. If this information is credible, it can reduce adverse selection by enabling consumers to avoid buying low-quality goods or paying less for poorer quality products.

For an outside organization to provide believable information, it must convince consumers that it is trustworthy. Consumers Union, which publishes the product evaluation guide *Consumer Reports*, tries to establish its trustworthiness by refusing to accept advertising or other payments from firms.

Unfortunately, experts undersupply information because it is a *public good* (nonrivalrous and only sometimes exclusive—see Chapter 17). Consumers Union does not capture the full value of its information through sales of its *Consumer Reports* because buyers lend their copies to friends, libraries stock it, and newspapers report on its findings. As a result, Consumers Union conducts less research than is socially optimal.

Auditing is another important example of third-party assessment, in which an independent accounting firm assesses the financial statements of a firm or other organization. Sometimes a firm obtains an audit voluntarily to enhance its reputation (a signal). Sometimes audits are required as a condition of being listed on a particular exchange or of participating in a particular transaction, and sometimes laws require audits be performed (screening).

Governments, consumer groups, industry groups, and others may establish a **standard**, which is a metric or scale for evaluating the quality of a particular product. For example, the R-value of insulation tells how effectively insulation works. Consumers learn of a brand's quality through **certification**: a report that a particular product meets or exceeds a given standard level.

Many industry groups set their own standards and get an outside group or firm, such as Underwriters Laboratories (UL) or Factory Mutual Engineering Corporation (FMEC), to certify that their products meet specified standard levels. For example, by setting standards for the size of the thread on a screw, we ensure that screws work in products regardless of brand.

When standard and certification programs inexpensively and completely inform consumers about the relative quality of all goods in a market and do not restrict the goods available, the programs are socially desirable. Licenses or certifications that restrict entry raise the average quality in the industry by eliminating low-quality goods and services. However, they drive up prices to consumers by restricting supply (see the Application "Occupational Licensing" in Chapter 2). As a result, welfare may go up or down, depending on whether the increased-quality effect or the higher-price effect dominates.

APPLICATION

Adverse Selection and Remanufactured Goods

Because consumers can't see a good before buying it over the Internet, it's easy for a shady seller to misrepresent its quality. In the worst-case lemons-market scenario, low-quality goods drive out high-quality goods.

Adverse selection concerns are particularly strong for electronic goods sold on eBay, as consumers cannot screen by inspecting the product before purchase. That this market exists on eBay indicates that sellers have found ways to signal quality or consumers have found ways to screen.

Consumers know that selling on eBay creates an enforceable contract, so sellers' signals reduce adverse selection on eBay. Sellers have a variety of ways to signal. Some sellers offer money back guarantees or warranties. Some pay extra to eBay to post flashy displays. One important signal is whether the good is new, remanufactured, or used. For years, manufacturers of cameras, computers, mobile phones, MP3 players, and other consumer durables have refurbished or upgraded returned products before trying to sell them again. Even though these remanufactured products may be comparable to new ones, consumers do not perceive them that way, so they are willing to pay a premium for new ones.

Neto et al. (forthcoming) studied sales on eBay of three types of iPods: the Classic, the Touch, and the Nano. They found, for example, that the average price of a used iPod Nano was 65% of a new one, while a remanufactured Nano was 82% of a new one. Thus, the signal of the good's type affects the price.

What about other signals such as positive descriptions? Consumers view used goods as varying more in quality than remanufactured or new products: the prices of used goods have greater variance than that of the others. As a result, quality claims may be more likely to affect the price of used goods than those of new and remanufactured goods. Neto et al. found that positive descriptions affect the price of most types of used iPods, but not new and remanufactured iPods.

Consumers can also screen using eBay's feedback (reputation) score: the percentage of positive ratings by past customers. A sleazy seller will have a bad feedback score. Subramanian and Subramanyam (2012) studied the sales of electronic goods on eBay. They found that a higher seller feedback score reduced the price differential between new and remanufactured goods. They also discovered that consumers pay higher prices for products remanufactured by the original manufacturer or their authorized factories than for those remanufactured by third parties.

Thus, third-party information in the form of ratings from previous customers, which consumers use to screen, and a variety of signals from firms help reduce the adverse selection problem for used and remanufactured goods.

Laws to Prevent Opportunism

In addition to setting standards and certifying goods and services, governments prevent opportunism in several ways. Three common examples are disclosure requirements, product liability laws, and universal coverage requirements.

Disclosure Requirements. Governments often require that the informed party disclose all relevant information to the uniformed party. For example, many local governments require that sellers disclose all relevant facts about a house to potential buyers such as its age and any known defects in the electrical work or plumbing. By doing so, these governments protect buyers against adverse selection due to undisclosed defects.

Product Liability Laws. In many countries, product liability laws protect consumers from being stuck with nonfunctional or dangerous products. Moreover, many U.S. state supreme courts have concluded that new products carry an implicit understanding that they will safely perform their intended functions. If they do not, consumers can sue the seller even in the absence of product liability laws. If consumers can rely on explicit or implicit product liability laws to force a manufacturer to compensate consumers for defective products, they do not need to worry about adverse selection. However, the transaction costs of going to court are very high.

Universal Coverage. Health insurance markets have adverse selection because low-risk consumers do not buy insurance at prices that reflect the average risk. Such adverse selection can be eliminated by providing insurance to everyone or by mandating that everyone buy insurance. Canada, the United Kingdom, and many other countries provide basic health insurance to all residents, financed by a combination of mandatory premiums and taxes. The U.S. Supreme Court confirmed the constitutionality of the "individual mandate" in the 2010 Patient Protection and Affordable Care Act, which requires virtually all Americans to have health care coverage.

Similarly, firms often provide mandatory health insurance as a benefit to all employees, rather than paying them a higher wage and allowing them to decide whether to buy such insurance on their own. By doing so, firms reduce adverse selection problems for their insurance carriers: Both healthy and unhealthy people are covered. As a result, firms can buy medical insurance for their workers at a lower cost per person than workers can obtain on their own.

18.3 Market Power from Price Ignorance

We've seen that consumer ignorance about quality can keep high-quality goods out of markets. We now illustrate that consumer ignorance about price variation across firms gives firms market power. As a result, firms have an incentive to make it difficult for consumers to collect pricing information. Because of this incentive, some stores won't quote prices over the phone.

If consumers (unlike sellers) do not know how prices vary across firms, *firms may gain market power and set prices above marginal cost*. Suppose that you go to Store A to buy a television set. If you know that Store B is charging $499 for the same TV, you are willing to pay Store A at most $499 (or perhaps a little more to avoid having to go to Store B). *Knowledge is power*. However, if you don't know Store B's price for that TV, Store A might charge you much more than $499. *Ignorance costs*.

In this section, we examine why asymmetric pricing information leads to noncompetitive pricing in a market that would otherwise be competitive. Suppose that many stores in a town sell the same good. If consumers have *full information* about prices, all stores charge the full-information competitive price, p^*. If one store raises its price above p^*, the store loses all its business. Each store faces a residual demand curve that is horizontal at the going market price and has no market power.

In contrast, if consumers have *limited information* about the price that firms charge for a product, one store can charge more than others and not lose all its customers. Customers who do not know that the product is available for less elsewhere buy from the high-price store.[5] Thus, each store faces a downward-sloping residual demand curve and has some market power.

Tourist-Trap Model

We now show that if such a market has a single price, it is higher than p^*. Suppose you arrive in a small town in California near the site of the discovery of gold. Souvenir shops crowd the street. Wandering by one of them, you see that it sells the town's

[5] A grave example concerns the ripping-off of the dying and their relatives. A cremation arranged through a memorial society—which typically charges a nominal enrollment fee of $10 to $25—often costs half or less than the same service when it is arranged through a mortuary. Consumers who know about memorial societies—which get competitive bids from mortuaries—can obtain a relatively low price.

they can break the monopoly-price equilibrium that occurs when consumers must search store by store for low prices. The more successful the advertising, the larger these stores grow and the lower the average price in the market. If enough consumers become informed, all stores may charge the low price. Thus, without advertising, no store may find it profitable to charge low prices, but with advertising, all stores may charge low prices.

18.4 Problems Arising from Ignorance When Hiring

Asymmetric information is frequently a problem in labor markets. Prospective employees may have less information about working conditions than firms do. Firms may have less information about potential employees' abilities than potential workers do.

Information asymmetries in labor markets lower welfare below the full-information level. Workers may signal and firms may screen to reduce the asymmetry in information about workers' abilities. Signaling and screening may raise or lower welfare, as we now consider.

Cheap Talk

Honesty is the best policy—when there is money in it. —Mark Twain

We now consider situations in which workers have more information about their ability than firms do. We look first at inexpensive signals sent by workers, then at expensive signals sent by workers, and finally at screening by firms.

When an informed person voluntarily provides information to an uninformed person, the informed person engages in **cheap talk**: unsubstantiated claims or statements (see Farrell and Rabin, 1996). People use cheap talk to distinguish themselves or their attributes at low cost. Even though informed people may lie when it suits them, it is often in their and everyone else's best interest for them to tell the truth. Nothing stops me from advertising that I have a chimpanzee for sale, but doing so serves no purpose if I actually want to sell a refrigerator. One advantage of cheap talk, if it is effective, is that it is a less-expensive method of signaling ability to a potential employer than paying to have that ability tested.

Suppose that a firm plans to hire Cyndi to do one of two jobs. The demanding job requires someone with high ability. The undemanding job can be performed better by someone with low ability because the job bores more able people, who then work poorly.

Cyndi knows whether her ability level is high or low, but the firm is unsure, initially thinking that either level is equally likely. Panel a of Table 18.1 shows the payoffs to Cyndi and the firm under various possibilities.[8] If Cyndi has high ability, she enjoys the demanding job: Her payoff is 3. If she has low ability, she finds the demanding job too stressful—her payoff is only 1—but she can handle the

[8]Previously, we used a 2×2 matrix to show a simultaneous-move game, in which both parties choose an action simultaneously. In contrast, in Table 18.1, only the firm can make a move. Cyndi does not take an action, because she cannot choose her ability level.

APPLICATION

Cheap Talk in eBay's Best Offer Market

In addition to auctions, eBay allows a seller to offer to sell a good for a specified price. The buyer may allow a potential buyer to respond with a *best offer* of a lower price. The seller may accept the best-offer bid, decline it, or make a counteroffer. The transaction is completed when a buyer or the seller accept the other side's offer.[9]

Backus et al. (2015) suggested that some sellers use cheap talk by posting an initial price that is a multiple of \$100. Items listed in multiples of \$100 receive offers 6 to 11 days sooner that are 5% to 8% lower, and are 3% to 5% more likely to sell than are items listed at similar "precise" prices such as \$99 or \$109. Thus, these round numbers may provide information that helps both parties: The seller makes a quick sale, and the customer buys at a low price.

Education as a Signal

No doubt you've been told that you should go to college to get a good job. Going to college may result in a better job because you obtain valuable training. Or, a college degree may land you a good job because it signals your ability to employers. If high-ability people are more likely to go to college than low-ability people, schooling signals ability to employers (Spence, 1974).

To illustrate how such signaling works, we'll make the extreme assumptions that graduating from an appropriate school serves as the signal and that schooling provides no useful training to firms (Stiglitz, 1975). High-ability workers are θ share of the workforce, and low-ability workers are $1 - \theta$ share. For a firm, the value of output produced by a high-ability worker is worth w_h, and that of a low-ability worker is w_l (over their careers). If competitive employers knew workers' ability levels, they would pay this value of the marginal product to each worker, so a high-ability worker receives w_h and a low-ability worker earns w_l.

However, suppose that employers cannot directly determine a worker's skill level. For example, when production is a group effort—such as in an assembly line—a firm cannot determine the productivity of a single employee.

Two types of equilibria are possible, depending on whether employers can distinguish high-ability workers from others. If employers have no way of distinguishing workers, the outcome is a **pooling equilibrium**: Dissimilar people are treated (paid) alike or behave alike. Employers pay all workers the average wage:

$$\overline{w} = \theta w_h + (1 - \theta)w_1. \tag{18.2}$$

Risk-neutral, competitive firms expect to break even because they underpay high-ability people by enough to offset the losses from overpaying low-ability workers.

We assume that high-ability individuals can get a degree by spending c to attend a school, but low-ability people cannot graduate from the school (or that the cost of doing so is prohibitively high). If high-ability people graduate and low-ability people do not, a degree is a signal of ability to employers. Given such a clear signal, the outcome is a **separating equilibrium**: One type of people takes actions (such as sending a signal) that allow them to be differentiated from other types of people. Here, a successful signal causes high-ability workers to receive w_h and others to receive w_l, so wages vary with ability.

[9]Because this game (Chapter 13) is very complicated, participants have difficulty devising optimal strategies.

signaling happens is that the private return to signaling, \$5,000, exceeds the net social return to signaling. The gross social return to the signal is zero because the signal changes only the distribution of wages. The net social return is negative because the signal is costly.

This inefficient expenditure on education is due to asymmetric information and the desire of high-ability workers to signal their ability. The government can increase total social wealth by banning wasteful signaling (eliminating schooling). Both low-ability and high-ability people benefit from such a ban.

In other cases, however, high-ability people do not want a ban. At point z (where $\theta = \frac{1}{2}$ and $c = \$5{,}000$), only a separating equilibrium is possible without government intervention. In this equilibrium, high-ability workers earn $w_h - c = \$35{,}000$ and low-ability workers make $w_l = \$20{,}000$. If the government bans signaling, both types of workers earn \$30,000 in the resulting pooling equilibrium, so high-ability workers are harmed, losing \$5,000 each. Thus, although the ban raises efficiency by eliminating wasteful signaling, high-ability workers oppose the ban.

In this example, efficiency can always be increased by banning signaling because signaling is unproductive. However, some signaling is socially efficient because it increases total output. Education may raise output because its signal results in a better matching of workers and jobs or because it provides useful training and serves as a signal. Also, education may make people better citizens. In conclusion, *total social output falls with signaling if signaling is socially unproductive but may rise with signaling if signaling also raises productivity or serves some other desirable purpose*.

Empirical evidence on the importance of signaling is mixed. For example, Tyler, Murnane, and Willett (2000) find that, for the least-skilled high school dropouts, passing the General Educational Development (GED) credential (the equivalent of a high school diploma) increases white dropouts' earnings by 10% to 19% but does not have a statistically significant effect on nonwhite dropouts.

Screening in Hiring

Firms screen prospective workers in many ways. An employer may hire someone based on a characteristic that the employer believes is correlated with ability, such as how a person dresses or speaks. Or a firm may use a test. Further, some employers engage in statistical discrimination, believing that an individual's gender, race, religion, or ethnicity is a proxy for ability.

Interviews and Tests. Most societies accept the use of interviews and tests by potential employers. Firms commonly assess abilities using interviews and tests. If such screening devices are accurate, firms benefit by selecting superior workers and assigning them to appropriate tasks. However, as with signaling, these costly activities are inefficient if they do not increase output. In the United States, the use of hiring tests may be challenged and rejected by the courts if the employer cannot demonstrate that the tests accurately measure skills or abilities required on the job.

Statistical Discrimination. If employers think that people of a certain age, gender, race, religion, or ethnicity have higher ability than others on average, they may engage in statistical discrimination (Aigner and Cain, 1977) and hire only people with that characteristic. Employers may engage in this practice even if they know that the correlation between these factors and ability is imperfect.[11]

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[11]Other common sources of employment discrimination are prejudice (Becker, 1971) and the exercise of monopsony power (Madden, 1973).

Figure 18.3 illustrates one employer's belief that members of Race 1 have, on average, lower ability than members of Race 2: Much of the distribution curve for Race 2 lies to the right of the curve for Race 1. Nonetheless, the figure also shows that the employer believes that the highest-skilled members of Race 1 have higher ability than the lowest-skilled members of Race 2: Part of the Race 1 curve lies to the right of part of the Race 2 curve. Still, because the employer believes that a group characteristic, race, is an (imperfect) indicator of individual ability, the employer hires only people of Race 2 if enough of them are available.

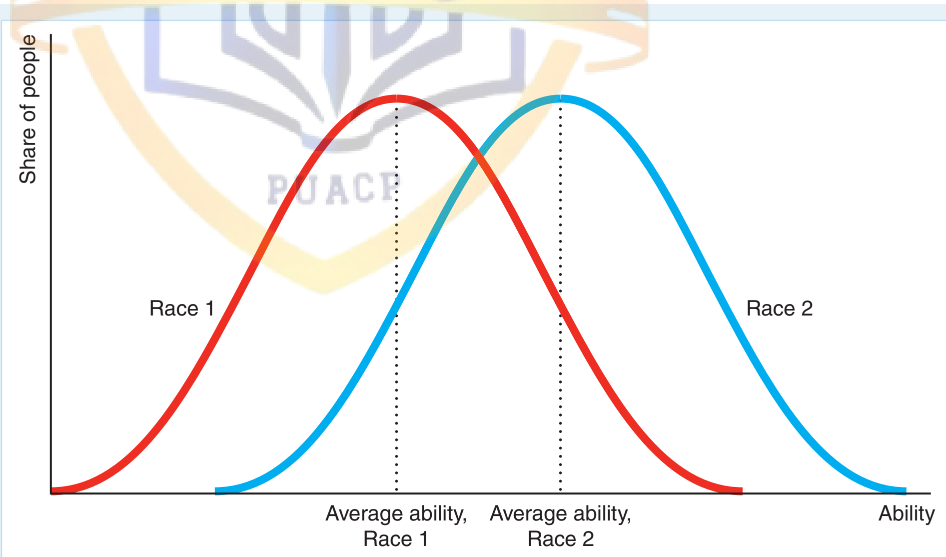
The employer may claim not to be prejudiced but to be concerned only with maximizing profit. Nonetheless, this employer's actions harm members of Race 1 as much as they would if they were due to racial hatred.

Even though ability distributions are identical across races, eliminating statistical discrimination is difficult. If all employers share the belief that members of Race 1 have such low ability that it is not worth hiring them, people of that race are never hired, so employers never learn that their beliefs are incorrect. Here, false beliefs can persist indefinitely. Such discrimination lowers social output by preventing skilled members of Race 1 from performing certain jobs.

However, statistical discrimination may be based on true differences between groups. For example, insurance companies offer lower auto insurance rates to young women than to young men because young men are more likely, *on average*, to have an accident. The companies report that this practice lowers their costs of providing insurance by reducing moral hazard. Nonetheless, this practice penalizes young men who are unusually safe drivers, and benefits young women who are unusually reckless drivers.

Figure 18.3 Statistical Discrimination

This figure shows the beliefs of an employer who thinks that people of Race 1 have less ability on average than people of Race 2. This employer hires only people of Race 2 even though the employer believes that some members of Race 1 have greater ability than some members of Race 2. Because this employer never employs members of Race 1, the employer may never learn that workers of both races have equal ability.



CHALLENGE SOLUTION

Dying to Work

In the Challenge at the beginning of the chapter, we asked whether a firm underinvests in safety if the firm knows how dangerous a job is but potential employees do not. Can the government intervene to improve this situation?

Consider an industry with two firms that are simultaneously deciding whether to make costly safety investments such as sprinkler systems in a plant or escape tunnels in a mine. Unlike the firms, potential employees do not know how safe it is to work